

TITANIC SURVIVAL PREDICTION USING LOGISTIC REGRESSION

1. Introduction

The objective of this project is to develop a machine learning model capable of predicting passenger survival on the Titanic using demographic and travel-related information. The project demonstrates the complete machine learning pipeline, including data preprocessing, exploratory data analysis, feature selection, model training, and performance evaluation.

2. Dataset Description

The Titanic dataset contains information about passengers aboard the Titanic.

Target Variable:

- * Survived

Selected Features:

- * Pclass

- * Sex

- * Age

- * SibSp

- * Parch
- * Fare
- * Embarked

Removed Features:

- * PassengerId
- * Name
- * Ticket
- * Cabin

3. Data Preprocessing

The following preprocessing steps were performed:

- * Missing values in Age were replaced with the median value.
- * Missing values in Embarked were replaced with the mode value.
- * Categorical variables were converted into numerical format.
- * Irrelevant columns were removed.

4. Exploratory Data Analysis

EDA revealed several important trends:

- * Female passengers showed higher survival rates.
- * First-class passengers had a greater chance of survival.
- * Age and fare displayed noticeable variation among survivors and non-survivors.

5. Model Development

Algorithm Used:

- * Logistic Regression

Train-Test Split:

- * Training Set: 80%
- * Testing Set: 20%

The model was trained using Scikit-learn's Logistic Regression implementation.

6. Performance Evaluation

Metrics Used:

- * Accuracy Score
- * F1 Score

- * Classification Report
- * Confusion Matrix

Results:

- * Accuracy: (Insert your result)
- * F1 Score: (Insert your result)

7. Limitations

- * Limited feature engineering.
- * Baseline model only.
- * No hyperparameter optimization.
- * Potential nonlinear relationships not captured.

8. Conclusion

The Logistic Regression model successfully classified passenger survival outcomes and provided a solid baseline for predictive modeling. The project demonstrated the complete machine learning workflow from data preprocessing to model evaluation.

9. Future Scope

- * Hyperparameter tuning.
- * Cross-validation.
- * Random Forest implementation.
- * Gradient Boosting methods.
- * Advanced feature engineering.
- * Ensemble learning techniques.